



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:



Course Name: B. Tech

Subject Name: Basics of Mechanical & Automation Engineering

Subject Code: 25ME101

Max. Marks: 15

Time: 75 Minutes

Semester: I

Note:

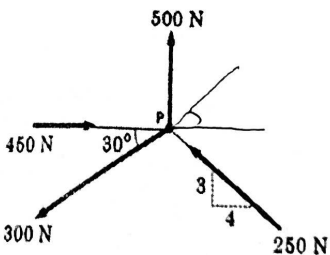
1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

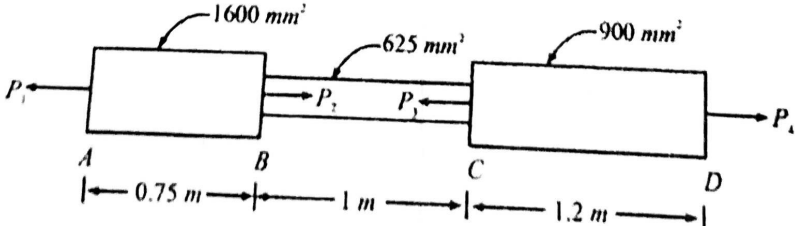
SECTION-A (Maximum Knowledge Level: K2)

Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 5)				
(a)	Write the conditions for equilibrium of a rigid body under coplanar concurrent forces.	CO1	K1	1.2.1
(b)	Explain the following: (a) Modulus of rigidity (b) Bulk modulus	CO1	K2	1.2.1
(c)	State the principle of superposition for deformation in a stepped bar.	CO1	K1	1.2.1
(d)	Name any four main components of an electric vehicle.	CO3	K1	1.2.1
(e)	Explain regenerative breaking in EVs/HEVs.	CO3	K2	1.2.1

SECTION-B (Minimum Knowledge Level: K2)

2. Attempt Any two Question (Same K Levels) (2 x 2)

(a)	With the help of a neat diagram, draw the stress-strain curve for mild steel and explain the salient features of each point on the curve.	CO1	K2	2.1.1
(b)	A system of four forces acting on a body is as shown in figure. Determine the magnitude resultant and its inclination with positive X-axis. 	CO1	K3	2.2.3

(c)	A member is subjected to point loads P_1 , P_2 , P_3 and P_4 as shown in figure. Calculate the force P_3 necessary for equilibrium if $P_1=110\text{kN}$, $P_2=210\text{kN}$ and $P_4=150\text{ kN}$. Determine change in length of the member. Take modulo of elasticity 210 GN/m^2 . 	CO1	K3	2.1.3
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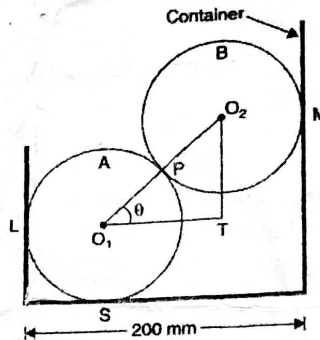
SECTION-C (Minimum Knowledge Level: K3)

(Same K Levels) (3 x 1)

3. Attempt Any one Question

(a)

Two spheres A and B weighing 100 N and 75 N respectively and with the corresponding radii 75 mm and 50 mm are placed in a container as shown in Figure. Determine the support reactions at points L, M, P and S. All surface contacts are frictionless.



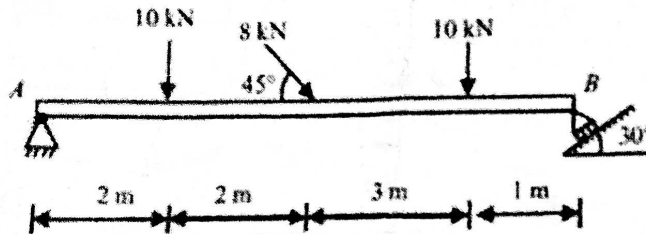
CO1

K3

2.2.1

(b)

Determine the support reactions at points A & B for the beam loaded as shown in figure. Neglect the thickness and weight of the beam.



CO1

K3

2.2.1

4. Attempt Any one Question

(Same K Levels) (3 x 1)

(a)

Discuss the working principle of an electric vehicle. What are the major demerits of these vehicles?

CO3

K2

2.1.1

(b)

What are series and parallel types of hybridization in HEVs?

CO3

K2

2.3.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

10/15



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)
Mid Term-II (Session: 2025-26)

Roll No.:

Course Name: B. Tech
Subject Name: Basics of Mechanical & Automation Engineering
Subject Code: 25ME101

Max. Marks: 15
Time: 75 Minutes
Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)

Q. No.	Question	CO	KL	PI
1. Attempt All Questions				
(1 x 5)				
① (a)	Explain the following terms associated with air conditioning; (i) Absolute Humidity, and (ii) Relative Humidity	CO2	K2	4.1.1,10.1.1
① (b)	Explain the following terms : (i) State, and (ii) Process	CO2	K2	4.1.1 10.1.1
① (c)	What is a reversible process?	CO2	K2	4.1.1,10.1.1
① (d)	What is scavenging?	CO3	K2	2.2.3, 2.4.3,3.2.1
① (e)	What is the purpose of a spark plug?	CO3	K2	2.2.3 2.4.3,3.2.1

SECTION-B (Minimum Knowledge Level: K2)

2. Attempt Any two Question				
(Same K Levels) (2 x 2)				
① (a)	With the help of P-h, T-s and block diagram, briefly explain the construction and working of VCRS systems.	CO2	K2	1.1.1 2.1.3 4.1.1
(b)	Explain the basic components and working of a window air conditioner with a suitable sketch.	CO2	K2	1.1.1 1.2.1
② (c)	Define a thermodynamic system. Differentiate between open system, closed system and an isolated system.	CO2	K2	4.1.1 10.1.1

SECTION-C (Minimum Knowledge Level: K3)

3. Attempt Any one Question **(Same K Levels) (3 x 1)**

(a)	The properties of a closed system change following the relation between pressure and volume as $pV = 3.0$ where p is in bar and V is in m^3 . Calculate the work done when the pressure increases from 1.5 bar to 7.5 bar.	CO2	K3	1.1.1 1.2.1 2.1.3
(b)	A cold storage is to be maintained at $-5^\circ C$ while the surrounding are at $35^\circ C$. The heat leakage from the surroundings into the cold storage is estimated to be 29 KW. The actual C.O.P. of the refrigeration plant is one-third of an ideal plant working between the same temperatures. Find the power required to drive the plant.	CO2	K3	1.1.1 1.2.1 2.1.3



4. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Explain the working of a four-stroke SI engine with neat and clean sketches.	CO3	K2	2.2.3 2.4.3 3.2.1
(b)	Describe the construction and operation of a two-stroke diesel engine with a neat and clear diagram.	CO3	K2	1.3.1 2.4.3 3.1.1

CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)
Pre - End Term Exam (Session: 2025-26)

Admission No.: [REDACTED]

Course Name: B. Tech.

Subject Name: Basics of Mechanical & Automation Engineering

Subject Code: 25ME101

Max. Marks: 20

Time: 90 Minutes

Semester: I

Note:

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SECTION-A (Knowledge Level: K2 & K3)

Q. No.	Question	CO	KL	PI
Attempt any one part from each question (3 Marks x 4 Questions = 12 Marks)				
1.(a)	Define sensor and transducer. Give two examples of each. Also, differentiate between active and passive transducers with examples.	CO4	K2	6.2.1
1.(b)	Explain the working and applications of train and ratchet mechanisms. How do they contribute to automation systems?	CO4	K2	6.2.1
2.(a)	Describe the different types of gears used in mechanical actuation systems. Discuss their specific applications.	CO4	K2	6.2.1
2.(b)	Write short notes on the following: (a) Direction control valve, (b) Pressure relief valve,	CO4	K2	6.2.1
3.(a)	Discuss in detail the construction and working of cylindrical robot.	CO5	K2	3.1.2
3.(b)	Explain robot specifications: DOF, payload capacity, reach, and why each is important for robot selection.	CO5	K2	3.1.2
4.(a)	Describe in detail the construction and working principle of Fused Deposition Modeling (FDM) with diagram.	CO5	K2	3.1.2
4.(b)	Describe in detail the construction and working principle of Selective Laser Sintering (SLS) with diagram.	CO5	K2	3.1.2

SECTION-B (Knowledge Level: K3)

Attempt any one part from each question (4 Marks x 2 Questions = 8 Marks)				
5.(a)	Explain the construction and working of a hydraulic actuation system with the help of a labeled diagram.	CO4	K2	6.2.1
5.(b)	Discuss the pneumatic actuation system in detail. Explain the purpose of the compressor, valves, and actuators in its operation.	CO4	K2	6.2.1
6.(a)	What is a SCARA robot? Explain its structure with diagram, DOF, working, advantages, limitations, and industrial applications.	CO5	K2	3.1.2
6.(b)	Explain the complete 3D printing workflow, starting from CAD modelling to postprocessing. Also discuss the major advantages and limitations of 3D printing compared to conventional manufacturing.	CO5	K2	3.1.2

CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create